

AI-Powered PDF Question Answering System Using Retrieval-Augmented Generation and Local LLM

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ABSTRACT:

The exponential growth of digital documents has created a strong demand for intelligent systems capable of extracting relevant information efficiently. Traditional keyword-based search techniques often fail to capture the semantic meaning of user queries, resulting in incomplete or irrelevant outputs. This paper presents an AI-powered PDF Question Answering System that leverages Retrieval-Augmented Generation (RAG) and a local Large Language Model (LLM) to deliver accurate and context-aware responses.

The proposed system enables users to upload PDF documents and interact with them through natural language queries. The system processes the document by extracting text, segmenting it into manageable chunks, and converting it into vector embeddings. These embeddings are stored in a FAISS vector database, enabling efficient semantic retrieval. Upon receiving a query, the system retrieves the most relevant content and generates responses using a locally deployed language model via Ollama.

The solution ensures data privacy, reduces dependency on cloud-based APIs, and provides faster response times for offline usage. Experimental results indicate improved performance over traditional search methods, making the system suitable for academic, technical, and research applications.

Keywords: Retrieval-Augmented Generation, PDF Question Answering, Large Language Model, FAISS, Ollama, Semantic Search.

I. INTRODUCTION:

In today's digital era, a vast amount of information is stored in document formats such as PDFs, including research papers, manuals, textbooks, and reports. Extracting relevant information from these documents is often time-consuming and inefficient when performed manually. Traditional search techniques rely on keyword matching, which lacks the ability to understand the context and meaning behind user queries.

With advancements in Artificial Intelligence and Natural Language Processing, more intelligent systems have been developed that can understand human language and provide meaningful responses. Large Language Models (LLMs) have demonstrated significant capabilities in text generation and comprehension. However, these models may produce inaccurate results when not grounded in specific data sources.

To overcome this limitation, Retrieval-Augmented Generation (RAG) combines information retrieval with generative models. This approach enhances accuracy by retrieving relevant information before generating a response. The proposed system utilizes RAG along with a locally deployed LLM to provide a secure, efficient, and accurate PDF-based question answering solution.

II. PROBLEM STATEMENT:

Extracting specific information from large PDF documents is a major challenge for students, researchers, and professionals. Manual reading is time-consuming, and traditional keyword-based search methods often fail to provide accurate results due to their inability to understand semantic relationships.

Cloud-based AI solutions offer improved capabilities but introduce concerns such as dependency on internet connectivity, high operational costs, latency, and potential data privacy risks. Therefore, there is a need for a system that can:

- * Understand natural language queries
- * Retrieve contextually relevant information
- * Generate accurate responses
- * Operate offline while ensuring data privacy

III. METHODOLOGY:

The proposed system follows a structured pipeline:

1. PDF Upload: Users upload a PDF file via the Streamlit interface.
2. Text Extraction: The system extracts textual data using a PDF processing library.
3. Text Chunking: The extracted content is divided into smaller chunks with overlap.
4. Embedding Generation: Each chunk is converted into vector embeddings using Sentence Transformers.
5. Vector Storage: Embeddings are stored in a FAISS vector database.
6. Query Processing: User queries are converted into embeddings.
7. Semantic Retrieval: Relevant chunks are retrieved using similarity search.
8. Response Generation: A local LLM generates answers based on retrieved context.

This pipeline ensures efficient and accurate information retrieval.

IV. SYSTEM ARCHITECTURE:

The system consists of multiple integrated components:

- User Interface (Streamlit)
- PDF Processing Module
- Text Chunking Module
- Embedding Generation Module
- Vector Database (FAISS)
- Retrieval Engine
- Local LLM (Ollama)
- Response Display Module

Workflow:

User → Upload PDF → Extract Text → Chunk Text → Generate Embeddings → Store in FAISS → Query → Retrieve Context → Generate Response → Display Output

V. TECHNOLOGIES USED:

Technology	Purpose
Python	Core programming language
Streamlit	Web interface
pypdf	Text extraction
LangChain	RAG pipeline
Sentence	Transformers Embedding
FAISS	Vector search
Ollama	Local model runtime
Phi/LLaMA	Language model

VI. IMPLEMENTATION:

The system is implemented using Python with a Streamlit-based frontend. The backend integrates LangChain for workflow orchestration, FAISS for vector storage, and Ollama for running local language models.

Caching techniques are applied to reduce repeated computations, improving system efficiency. Lightweight models such as Phi are used to ensure faster response times even on moderate hardware configurations.

VII. RESULTS AND DISCUSSIONS:

The system was tested using academic and technical PDF documents.

The results show that:

- * The system provides accurate and context-aware answers
- * Semantic retrieval outperforms keyword-based search
- * Local LLM ensures fast and private responses
- * The chatbot interface improves user interaction

Example queries successfully handled:

- * Define data structures
- * Explain stack and queue
- * Describe algorithm complexity

VIII. ADVANTAGES:

- * Improved accuracy through semantic understanding
- * Offline functionality with local LLM
- * Data privacy and security
- * Reduced dependency on cloud services
- * Interactive chatbot interface
- * Faster information retrieval

IX. LIMITATIONS:

- * Performance depends on hardware capabilities
- * Large PDFs may increase processing time
- * Accuracy depends on text extraction quality
- * Limited performance with highly complex queries

X. FUTURE SCOPE:

Future improvements may include:

- * Multi-document support
- * Voice-based interaction
- * Multilingual support
- * Cloud deployment
- * GPU acceleration
- * Support for additional file formats

XI. CONCLUSION:

The AI-Powered PDF Question Answering System successfully integrates Retrieval-Augmented Generation with a local Large Language Model to provide an efficient and intelligent document interaction platform. The system addresses the limitations of traditional search methods by offering context-aware responses while maintaining privacy through offline execution.

This project demonstrates the potential of combining semantic search with generative AI for real-world applications in education, research, and industry.

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